ENGR 312: Automatic Control Systems

Homework Assignment #6

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Problem 1

1(a) $2\dddot{y} + 3\ddot{y} + 5\dot{y} + 2y = r(t)$

$$\dddot{y} + 1.5\,\ddot{y} + 2.5\,\dot{y} + y = 0.5\,r$$

$$x_1 = y, \qquad x_2 = \dot{y}, \qquad x_3 = \ddot{y}$$

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= x_3\\ \dot{x}_3 &= -x_1 - 2.5\,x_2 - 1.5\,x_3 + 0.5\,r\end{aligned}$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -2.5 & -1.5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 0.5\end{bmatrix} r, \qquad y = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

1(b) $y^{(4)} + 1.5\ \dddot{y} + 2.5\,\dot{y} + y = 2r(t)$

$$x_1 = y, \qquad x_2 = \dot{y}, \qquad x_3 = \ddot{y}, \qquad x_4 = \dddot{y}$$

$$\begin{aligned}\dot{x}_1 &= x_2\\ \dot{x}_2 &= x_3\\ \dot{x}_3 &= x_4\\ \dot{x}_4 &= -x_1 - 2.5\,x_2 - 0\cdot x_3 - 1.5\,x_4 + 2\,r\end{aligned}$$

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\\ \dot{x}_4\end{bmatrix} = \begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\\ -1 & -2.5 & 0 & -1.5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 0\\ 2\end{bmatrix} r, \qquad y = \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\\ x_4\end{bmatrix}$$

Problem 2

2(a) $\dfrac{Y(s)}{U(s)} = \dfrac{s^2 + 3s + 7}{(s+1)(s^2 + 5s + 4)}$

$$(s+1)(s^2 + 5s + 4) = s^3 + 6s^2 + 9s + 4$$

```
num = [1 3 7];
den = conv([1 1], [1 5 4]);
[A, B, C, D] = tf2ss(num, den)
```

```
A =
    -6    -9    -4
     1     0     0
     0     1     0
B =
     1
     0
     0
C =
     1     3     7
D =
     0
```

$$\dot{\mathbf{x}} = \begin{bmatrix} -6 & -9 & -4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u, \qquad y = \begin{bmatrix} 1 & 3 & 7 \end{bmatrix} \mathbf{x}$$

2(b) $\dfrac{Y(s)}{U(s)} = \dfrac{-25}{s^2 + 4s + 25}$

```
num = -25;
den = [1 4 25];
[A, B, C, D] = tf2ss(num, den)
```

```
A =
    -4   -25
     1     0
B =
     1
     0
C =
     0   -25
D =
     0
```

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & -25 \\ 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u, \qquad y = \begin{bmatrix} 0 & -25 \end{bmatrix} \mathbf{x}$$

Problem 3

3(a)

$$\mathbf{A} = \begin{bmatrix} -1 & 0 & 1 \\ 1 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}, \quad D = 0$$

```
A = [-1 0 1; 1 -2 0; 0 0 -3];
B = [0; 0; 1];
C = [1 1 0];
D = 0;
[num, den] = ss2tf(A, B, C, D)
G = tf(num, den)
Gmin = minreal(G)
```

```
num =
         0         0    1.0000    3.0000
den =
     1     6    11     6
G =
          s + 3
  ----------------------
  s^3 + 6 s^2 + 11 s + 6

Continuous-time transfer function.
Gmin =
        1
  -------------
  s^2 + 3 s + 2

Continuous-time transfer function.
```

$$G(s) = \frac{s+3}{s^3+6s^2+11s+6} = \frac{s+3}{(s+1)(s+2)(s+3)} = \frac{1}{s^2+3s+2}$$

$$X_3 = \frac{U}{s+3}, \qquad X_1 = \frac{X_3}{s+1}, \qquad X_2 = \frac{X_1}{s+2}$$

$$Y = X_1 + X_2 = X_1 \frac{s+3}{s+2} = \frac{U}{(s+3)(s+1)} \cdot \frac{s+3}{s+2} = \frac{U}{(s+1)(s+2)}$$

3(b)

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad D = 0$$

```
A = [1 2; -4 -3];
B = [1; 2];
C = [1 1];
D = 0;
[num, den] = ss2tf(A, B, C, D)
G = tf(num, den)
```

```
num =
         0    3.0000    1.0000
den =
     1     2     5
G =
     3 s + 1
  -------------
  s^2 + 2 s + 5

Continuous-time transfer function.
```

$$G(s) = \frac{3s+1}{s^2+2s+5}$$

$$\det(s\mathbf{I}-\mathbf{A}) = (s-1)(s+3)-(2)(-4) = s^2+2s+5$$

$$(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B} = \frac{1}{s^2+2s+5}\begin{bmatrix} s+3 & 2 \\ -4 & s-1 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{s^2+2s+5}\begin{bmatrix} s+7 \\ 2s-6 \end{bmatrix}$$

$$\mathbf{C}(s\mathbf{I}-\mathbf{A})^{-1}\mathbf{B} = \frac{(s+7)+(2s-6)}{s^2+2s+5} = \frac{3s+1}{s^2+2s+5}$$